Public Release R0

The Realignment Architecture

A public research prospectus for evidence-bound resilience systems

William Edgar Brissey

Independent AI Systems Architect, Developer & Consultant

EVIDENCE ▪ PROVENANCE ▪ FALSIFICATION ▪ HUMAN AUTHORITY

27 August 2026

Research prospectus—hypotheses, architecture, and evaluation plan

A controlled public prospectus that defines falsifiable questions, evidence standards, break conditions, evaluation work packages, and governance boundaries for a cross-domain resilience research program.

Table of contents

- **Status:** Public Release R0
- **Revision:** R0
- **Claim classification:** Research prospectus—hypotheses, architecture, and evaluation plan
- **Role:** Independent AI Systems Architect, Developer & Consultant

1 Executive purpose

The Realignment Architecture is a research program for examining how environmental stress, infrastructure fragility, computational dependence, maintenance capacity, and institutional decision-making may interact under uncertainty.

This document converts an earlier architecture white paper into a public research prospectus. It preserves the useful systems questions while withdrawing any implication that a specific collapse timeline, causal chain, physical mechanism, or resilience design has already been proven.

The program is ambitious by design, but its claims are provisional. Its value will depend on whether independent reviewers can inspect the evidence, reproduce the analysis, identify contradictions, and reject unsupported hypotheses.

2 What this prospectus does not claim

This release does not claim that:

- a particular climate or infrastructure collapse date has been established;
- Atlantic, Pacific, geophysical, economic, or biological processes form one verified causal chain;
- a proposed resilience node is technically, economically, environmentally, or legally feasible;
- an AI architecture can replace expert judgment or accountable human authority;
- equations, simulations, or model agreement constitute empirical validation; or
- a complete engineering blueprint or deployable product is being released.

Claims involving climate dynamics, ocean circulation, geophysics, public safety, nuclear systems, biology, infrastructure, economics, or regulation require domain-specific review and primary evidence.

3 Research objective

The objective is to determine whether a provenance-first systems architecture can improve the quality of cross-domain resilience research by making five things explicit:

1. what was observed;
2. what was inferred;
3. what remains a hypothesis;
4. what would falsify the hypothesis; and
5. what human decision, if any, the evidence can responsibly support.

The architecture is therefore a method for organizing inquiry before it is a theory of the world.

4 Core research questions

4.1 RQ1 — Coupled stress

Under what measurable conditions do stresses in environmental, infrastructure, computational, economic, and human-maintenance systems amplify one another rather than remain independent?

4.2 RQ2 — Graceful degradation

Which decentralized architectures provide demonstrably better continuity, recoverability, and failure containment than comparable centralized designs under defined stress scenarios?

4.3 RQ3 — Evidence-bound AI

Can a human-gated AI workflow preserve provenance, disagreement, uncertainty, and revision history better than an ordinary document- and dashboard-centered research workflow?

4.4 RQ4 — Local resilience feasibility

Can proposed local resilience systems be described with complete energy, water, material, labor, maintenance, safety, regulatory, and economic balance sheets before implementation claims are made?

4.5 RQ5 — Inter-basin hypotheses

Which proposed links among oceanic and atmospheric processes are supported by primary observations, which are only correlations, and which fail when tested against alternative explanations?

5 Hypothesis register

ID	Provisional hypothesis	Required evidence	Break condition
H1	Cross-domain stress can create nonlinear service degradation.	Time-aligned, versioned observations across defined system boundaries and comparison against simpler independent-stress models.	A simpler model explains the observations equally well or better.
H2	Decentralized systems can improve continuity under selected failure modes.	Comparable architectures, explicit loads, failure injections, recovery metrics, and lifecycle costs.	Benefits disappear after maintenance, coordination, security, or cost burdens are included.

ID	Provisional hypothesis	Required evidence	Break condition
H3	Provenance-first, human-gated AI can improve auditability.	Controlled workflow comparisons measuring citation accuracy, evidence recovery, contradiction detection, reviewer effort, and error rates.	The architecture fails to improve predefined measures or introduces unacceptable new failure modes.
H4	A local resilience-node concept can be evaluated before construction.	Conserved-unit balance sheets, supplier and maintenance assumptions, hazard analysis, regulatory review, and sensitivity analysis.	Any critical input, waste stream, safety function, or maintenance dependency cannot be closed or responsibly bounded.
H5	Selected inter-basin relationships may be distinguishable from shared forcing or coincidence.	Primary datasets, lag specifications, causal assumptions, null models, out-of-sample tests, and independent replication.	Results are unstable across datasets, preprocessing choices, time windows, or credible null models.

The register is a starting point. Each material claim must eventually have an owner, source set, test, status, and revision history.

6 Architecture of the research program

6.1 1. Realignment evidence layer

The Realignment evidence layer is a proposed method for compiling sources, datasets, claims, counterclaims, assumptions, and unresolved contradictions into an inspectable research graph.

Its intended output is not automatic truth. It is a traceable record showing how a conclusion was assembled and where the record remains weak.

6.2 2. Second Mind

The Second Mind is a local-first, human-reviewed advisory architecture. It is intended to support resumable research, explicit uncertainty, and recoverable decision trails.

It must not make unreviewed high-consequence decisions, conceal model disagreement, or convert probabilistic output into authoritative instruction.

6.3 3. Agent-trace diagnostics

The most immediate engineering work is an evidence-bound, post-mortem agent-trace diagnostic platform. It is intended to examine supplied workflow traces, identify the earliest observed evidence-backed divergence, bind findings to exact evidence, and preserve human adjudication.

This component provides the most direct near-term opportunity for measurable software evaluation.

6.4 4. EDEN pre-feasibility

EDEN is a pre-feasibility concept for localized resilience. Any public technical treatment must begin with conserved quantities, dependency inventories, failure modes, lifecycle maintenance, safety cases, regulatory constraints, and economic sensitivity.

A concept diagram is not a feasibility result.

6.5 5. Inter-Basin Coupling work stream

Inter-Basin Coupling is an exploratory research stream. It will separate observation from correlation, correlation from causal inference, and causal inference from forecast.

No inter-basin claim should advance without primary data provenance, explicit lag choices, alternative hypotheses, sensitivity analysis, and a reproducible test plan.

7 Claim discipline

Every substantive statement should be labeled internally using the following classes:

Class	Meaning	Publication requirement
Observation	Directly reported or measured information.	Primary source, dataset identifier, date, unit, and retrieval path.
Inference	A conclusion drawn from observations.	Stated reasoning, assumptions, uncertainty, and alternatives.
Hypothesis	A testable provisional explanation.	Falsification test, break condition, and status.
Design proposal	A candidate architecture or mechanism.	Interfaces, dependencies, hazards, validation plan, and maturity.
Decision	A human-authorized action.	Accountable owner, evidence reviewed, limits, and revision record.

Rhetorical confidence must never substitute for movement from one class to another.

8 Evaluation program

8.1 Work package 1 — Source and claim audit

- build a primary-source inventory;
- record dataset and document versions;
- identify unsupported numerical and causal claims;
- separate historical observation from forward projection; and
- preserve credible disagreement.

8.2 Work package 2 — Competing-model tests

- define system boundaries and measurable variables;
- test proposed relationships against simpler explanations;
- declare lag choices and preprocessing decisions;
- use out-of-sample evaluation where possible; and
- publish null and negative results.

8.3 Work package 3 — Agent-trace diagnostic prototype

- define representative trace formats;
- create synthetic and public test cases;
- measure evidence binding, divergence localization, false positives, false negatives, and reviewer effort;
- red-team prompt injection and provenance failures; and
- retain human decision authority.

8.4 Work package 4 — EDEN balance sheets

- quantify energy, water, material, waste, labor, maintenance, and replacement flows;
- identify single points of failure and external dependencies;
- conduct hazard and regulatory reviews;
- compare alternatives; and
- state clearly when a boundary cannot be closed.

8.5 Work package 5 — Governance and release controls

- maintain revisioned claim registers;
- document reviewer disagreement;
- define stop-work and withdrawal criteria;
- distinguish research releases from product claims; and
- publish corrections against a controlling version.

9 Minimum evidence standard

A material claim should not appear in a controlling release unless the record identifies:

- the primary source or dataset;
- the exact observation used;
- the unit, time range, and system boundary;
- the transformation or analysis applied;
- the uncertainty and competing interpretation;
- the test that could disconfirm the claim; and
- the revision that introduced or changed it.

Secondary synthesis can help discover sources, but it cannot replace inspection of the underlying evidence.

10 Safety and governance boundaries

The program is advisory and research-oriented. It does not authorize autonomous control of infrastructure, medical care, financial systems, public safety, energy facilities, or other high-consequence operations.

Human review is not a ceremonial final step. It is part of the architecture. Reviewers must be able to reject a model conclusion, inspect its evidence trail, and record a reason without the system rewriting the disagreement as consensus.

Sensitive data, proprietary traces, and personal information require explicit handling rules, access controls, minimization, and deletion procedures before evaluation.

11 Publication sequence

The controlling prospectus will remain available as one complete document. Subsequent releases will be modular and will link back to it:

1. **Foundations and claim discipline**
2. **Environmental and inter-basin hypotheses**
3. **Infrastructure stress and graceful degradation**
4. **Second Mind and agent-trace diagnostics**
5. **EDEN pre-feasibility and conserved-unit balance sheets**
6. **Governance, economics, falsification, and release controls**

Modules may be released chapter by chapter when they can stand independently. Longer technical groupings may be released as books only when all included chapters share one evidence standard and revision state.

No module inherits credibility merely because it appears in the same architecture.

12 Collaboration request

I welcome researchers, systems engineers, software evaluators, domain specialists, independent builders, and critical reviewers who can:

- test or falsify a specific hypothesis;
- identify missing primary sources;
- propose competing explanations;
- review safety, feasibility, regulatory, or economic assumptions;
- contribute reproducible tests or public datasets; and
- help define measurable software-evaluation criteria.

The immediate collaboration priority is the agent-trace diagnostic prototype because it can be evaluated with bounded public or synthetic data while the broader physical and resilience claims undergo slower domain review.

13 Revision and withdrawal policy

A controlling release will be revised when stronger evidence changes a claim, boundary, or test. A claim will be downgraded or withdrawn when its evidence cannot be reproduced, when a credible alternative performs better, or when required provenance is unavailable.

Older revisions may remain in repository history for provenance, but they must not be presented as current.

For the program-wide publication principles, see Status and Purpose of This Work.

Reality supersedes theory.